

Total No. of Questions : 10]

SEAT No. :

P2339

[Total No. of Pages : 2

[5254]-674

B.E. (Computer Engineering)

MOBILE COMPUTING

(2012 Pattern) (End Semester) (Elective - III)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary*

- Q1)** a) Explain IG,2G,3G and 4G [5]
b) What is the need of spread spectrum technology? Explain Slow FHSS.[5]

OR

- Q2)** a) Explain GSM Network Architecture. [5]
b) What are the limitations of Mobile Devices? [5]

- Q3)** a) Explain DSSS with one example. [5]
b) Draw and explain Mobile Computing architecture for a mobile device.[5]

OR

- Q4)** a) Explain various traffic and control data channels used in GSM network. [5]
b) Explain how the confidentiality is maintained during a call using ciphering and encryption processes? [5]

P.T.O.

- Q5)** a) Describe the registration of a visiting mobile node in handover. How is the binding between the home agent and the foreign agent created? [6]
b) Describe Indirect TCP. Explain the modifications of Indirect TCP as the selective repeat protocol. [6]
c) Explain Cluster-head Gateway Switch Routing (CGSR) in MANET. [6]

OR

- Q6)** a) Explain what is MANET. What are the security threats to a MANET? [6]
b) Explain the role of Agent Discovery, Agent advertisement and Agent Solicitation during location Management in GSM Network. [6]
c) List the entities of mobile IP and describe data transfer from a mobile node to a fixed node and vice-versa. [6]

- Q7)** a) Describe pull-based data-delivery mechanism. What are the advantages and disadvantages of pull-based data-delivery? [8]
b) Explain Data Synchronization Protocols for synchronizing mobile applications at mobile devices. [8]

OR

- Q8)** a) Show architecture for data dissemination and broadcast. Explain the reason for communication asymmetry in mobile network. [8]
b) What are the different types of data synchronizations in mobile computing systems? Describe synchronization usage models in mobile applications. [8]

- Q9)** a) Explain the role of a gateway in connecting networks using different protocols. Describe a transcoding gateway and its applications in mobile computing systems. [8]
b) Write Short note on:
i) Android OS
ii) Service discovery and device management. [8]

OR

- Q10)** a) Describe the functions of a mobile agent. Why does an agent move from tier to tier during an application? [8]
b) Write Short note on: [8]
i) Mobile-agent-based architecture
ii) IOS

